

Pick and Place Industrial Robot Controller with Computer Vision

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Abstract—In these days, most of the robots operate in a pre-coded cycle, to the best, the cycle waits for a few input sensors to further along cycle. When we compare this with the way a human being does the same job, we have a lot more flexibility as the path to be taken by our hand is formulated for each cycle depending on the environment under robot. This paper gives an idea about finding the x, y coordinate of object. This is found by converting pixel co-ordinate into real world co-ordinate with the help of 2D transformation. We incorporate robotic arm controller based on location and orientation of object. Fundamentally the problem taken in this work is to pick and place objects which are placed under vision sensor. Computer vision is the visual discernment segment of an eager motivation to copy human insight and to enrich robots with savvy conduct. This work has been carried under the guidance of Castalia Research lab, Pune.

Keywords— computer vision, 2D transformation, pick and place, pixel co-ordinate, world co-ordinate.

I. INTRODUCTION

Nowadays people are superseded by robot in performing risky and dangerous task which are not done due to size, environment constraints. The juncture of a few propelled advances is bringing the period of mechanical autonomy ever closer. Today's industrial robots are more diminutive, more frugal, more practical and cost-efficacious. Pune is hub of industrial robot, for example Tata's Pune plant alone has invested in 100 robots [1]. The industrial robots include ABB, Kuka, Funuc, Yaskawa Motoman etc [1]. We carried our work on ABB robot. In India ABB robot's demonstration center is in Pune, Bangalore, Chennai. Castalia research lab provides internship for 3D computer vision and machine vision based robotics. Machine vision is vast area of interest and is related to computer vision.

Machine vision has better sensitivity and resolution. It tends to application for identifying and inspecting small parts, guiding the robot controllers. If we look at automation industry, vision is most important. It can be used for number of industry application such as welding, painting, automatic pick and place object, military operation, in space, in underwater operation, in agriculture operation etc. The most expensive and time consuming task is to place objects in specific manner known as palletizer. Main contribution of robotics is in palletising. These objects are random and oriented, vision

system has to indentify objects and arrange in order. The requirements of industries like Mahindra, Volkswagen is pick and place the random products.

Robot is programmable virtual artificial agent, and general purpose electromagnetic machine that have some characteristics like human and that is guided by computer program. Pick and place object is major task in industry environment [2]. Many methods have been proposed for vision based grasping control. While grasping tasks, robot needs earlier knowledge of scene analysis [3]. It requires to undertake two steps such as object reconstruction and grasp planning. Object to be grasp needs technique such as fuzzy logic [3], neural network, image and position based visual servoing [2] etc. The idea behind the work is grasping and interaction with objects. The motivation behind this work is vision based grasp planning and vision based pick and place control.

The objective of these visual servoing approach is control 6 degree of freedom robot with help of information provided by vision sensor. The whole work is divided into 4 subprocess. The first is scene analysis, which consists of object position and orientation calculation. The second is co-ordinate conversion between camera co-ordinate and robot co-ordinate. The third is path planning based on vision system. The last is interfacing vision system to robot, which consists of ethernet and socket communication.

II. RELATED WORK

The presented work in this paper is related machine learning and robotics. The entire work is isolated into vision processing, co-ordinate transformation, position calculation, orientation algorithm and guideline unraveling to robot.

Boundary detection has huge applications in areas such as segmentation, contour grouping and symmetry detection [4]. Image segmentation is based on colour, intensity and texture. It may be region based segmentation, feature based, edge detection, thresholding based or model based [5]. The uncalibrated view of the plane can be converted into calibrated plane using transformation properties. The change in the rotation, size, shape alters the object's coordinate [7]. There is proposed planner that investigates camera state space and robot space [8].

2D transformation is utilized while mapping pixel co-ordinate into world co-ordinate. Transformations are acclimated to reposition objects, to resize the objects, to transmute graphics of screen. The generalized boundary detection method (Gb) simultaneously cumulates low-level and mid-level image representations in a single eigenvalue quandary [4]. This is utilized to disentangle the configuration of visual servo controller. Visual servoing is nothing but coalescence of vision framework and robot controller. According to International Standards Organization Industrial robots are classified as automatically controlled, reprogrammable, multipurpose manipulator programmable in three or more axes [9]. Yaskawa Motoman term them S, L, U, R, B and T for axis 1 to 6 DOF respectively. Fanuc named them as J1 to J6. During grasp planning, object selection is based on goal directed manner which involves Gaussian mixture models and probabilistic graphical model that have conditional dependencies such as Bayesian networks [6]. Contingent upon this it needs to perform control and way path orchestrating. The another grasp planning algorithm consists of virtual elastic reconstruction algorithm [3]. The surface psychologists toward the article until a few focuses capture the item visual body. These strengths are repaid by creating powers to the visual body[3].

III. SCENARIO DESCRIPTION

This area depicts situation consider in this paper. A robot aims to grasp coordinate of rectangular objects and put them into box according to their weights. We first present setup for this work, system framework for robot manipulator with some assumption.

A. Experimental Setup

Experimental setup consists 6 DOF robot in front of a working table, where rectangular objects are put. Camera is placed above the table, from where all objects are noticed and connected to machine program. We are going to sort the object on the substratum of area of object. Machine programming is done with help of OpenCV library. Machine programming includes object recognition, image preprocessing part, conversion of pixel co-ordinate into world co-ordinate using transformation properties and orientation calculation of object so that it will guide the robot through how much angle it has to rotate. Fig. 1 shows 6 DOF robot manipulator [9].

B. Assumptions

The experimental setup depends on visual detecting information and a few presumptions. Following postulations are noted while considering the system framework.

The surface on which we put the item is uniform. The stature of surface is uniform. The optical pivot of camera is perpendicular to working table and camera is well calibrated. The camera does not move during work.

C. System Framework

Fig. 2 shows the system block diagram which consists of vision sensor, mapping of camera coordinate into world coordinate, orientation calculation algorithm and inverse kinematics controller.

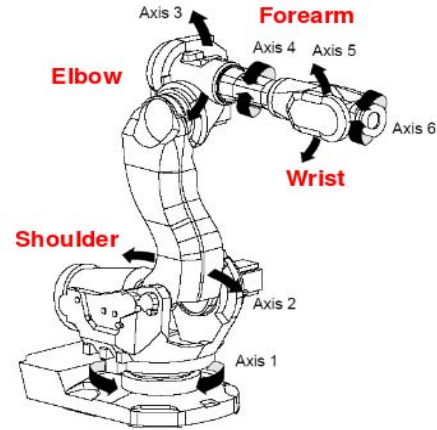


Fig. 1. 6 DOF robot

Here we utilized stand out camera, mounted over the area of interest which is utilized of object recognition, preprocessing steps. This can have another advantage as it avails for inspection of object, which eliminates desideratum of second camera. Section IV describes world coordinate conversion for position and orientation estimation algorithm. This converted coordinate will be utilized as positions in world coordinate of objects.

IV. OBJECT POSITION AND ORIENTATION ESTIMATION

The vision processing is carried by OpenCV library which fixates on real time computer vision. It is vision library for Robot Operating System and provides interfaces for C, C++, python on Windows, Linux, MacOS, iOS, FreeBSD, NetBSD, OpenBSD and Android [10]. We have used Python 2.7.6 having libraries like Numpy, matplotlib, Tkinter etc.

Transformation has properties like translation, scaling, rotation, affine and shearing. To map camera coordinate into world coordinate, 2x2 matrix can be converted into 3x3 matrix by homogeneous coordinate.

Let translation distances are t_x and t_y , we add this terms to the pristine coordinate position (x, y) to move the point to an incipient position (x', y') [7]. (t_x, t_y) is known as translational vector or shift vector.

Translated position can be written as below

$$x' = x + t_x \quad (1)$$

$$y' = y + t_y \quad (2)$$

This can be written as

$$\begin{bmatrix} x' \\ y' \end{bmatrix} = \begin{bmatrix} x \\ y \end{bmatrix} + \begin{bmatrix} t_x \\ t_y \end{bmatrix} \quad (3)$$

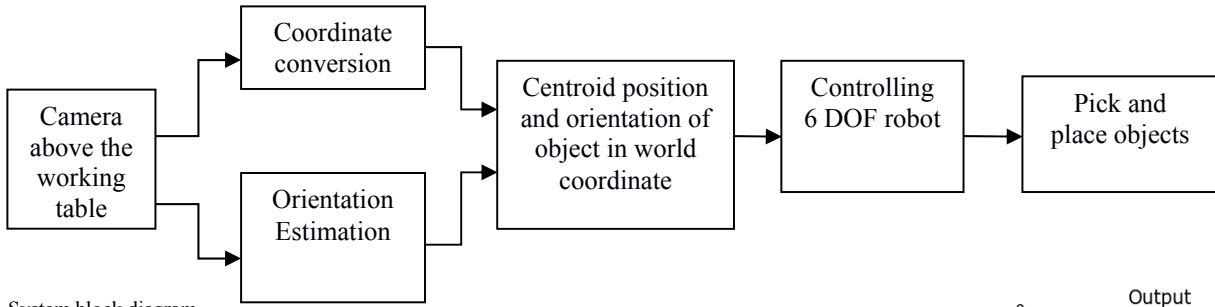


Fig. 2. System block diagram

The repositioning object along with circular xy plane rotation of object is applied. θ be the rotational angle, (x, y) be the position of points to rotate and (x', y') are the position of rotated point.

$$\begin{bmatrix} x' \\ y' \end{bmatrix} = \begin{bmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} \quad (4)$$

The scaling transformation modifies the size. We have done experiment on polygon by multiplying the coordinate values (x, y) of each vertex by scaling factors s_x and s_y , to produce the transformed coordinates (x', y') . The result is shown in fig. 3 and $s_x = s_y = 2$.

$$\begin{bmatrix} x' \\ y' \end{bmatrix} = \begin{bmatrix} s_x & 0 \\ 0 & s_y \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} \quad (5)$$

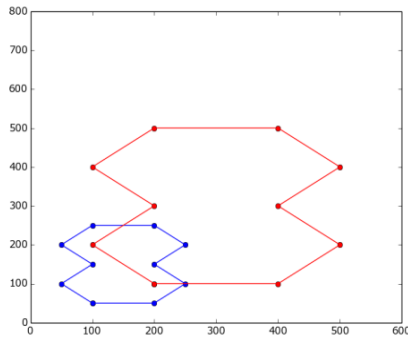


Fig.3. Scaled transformation of polygon

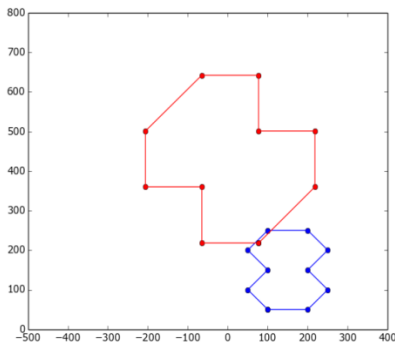


Fig. 4. Translation, rotation and scaling transformation of polygon

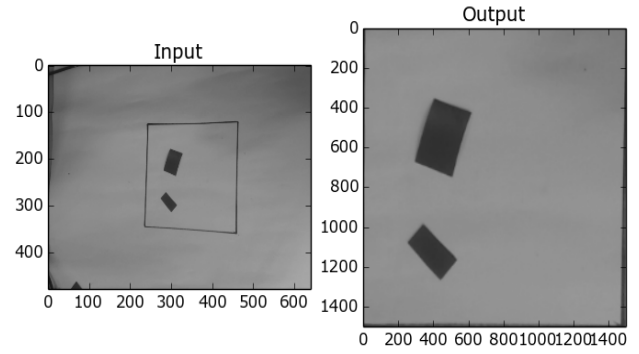


Fig. 5. Extracted region of interest

The translation, rotation and scaling of polygon with $s_x = 2$, $s_y = 2$, $t_x = 3$, $t_y = 3$ and $\theta = 45$ degree is shown in fig 5. By using above transformation properties we had extracted region of interest and scaled according to our need. This is shown in fig. 5. Thus it forms the world co-ordinate from camera co-ordinate. Now next step is object recognition. This is done by OpenCV Python programming. In OpenCV edge detection is known as counter detection. Here we performed the sorting based on area of contour. According to X scale and Y scale we get the coordinate of centroid. The result is shown in fig. 6. Each joint angle of robot can rotate with respective of its maximum angle and move. Vision system is interfaced with robot by openabb. After getting world co-ordinate of rectangular objects and its orientation, robot will pick the objects and place into suitable box. Now the communication between vision system and controller is done through Ethernet. The ethernet switch is connected to robot controller camera and through GigE to vision processor. GigE is interface standard for camera and gives a system to transmitting fast video and control information over ethernet systems. We used socket communication to pass object coordinates to the controller. The advantage of socket is there is no need to keep vision system near robot. We can pass coordinates from remote distance. Vision system sends 'set_cartesian, x, y, theta' to the controller IRC5. IRB 120 ABB robot is equipped with compact IRC5 controller. This controller supports client server socket communication. This controller is able to parse string with "," as delimiter. So that controller shall execute a movement of robot to the position given in string. Again it is possible to 'get cartesian' of robot

current position to controller. This controller shall send back current position of robot as a string 'x, y, theta'.

As the scene analysis is completed second stage is planning process of gripper controller. We have constructed orientation algorithm by minimum area of boundary box around the object in OpenCV. According to orientation of object gripper will take action to pick the object. Objects are sorted depending on their area. If area of object is greater than 1000 cm x cm, robot will place in one box. If area of object is less than 1000 cm x cm, robot will place in another box. We have used vacume gripper as it provides good handling of object if surface is smooth, uniform and flat. Vacuum grippers are utilizable in the cumbersomely hefty industries.

V. EXPERIMENTAL RESULTS

Fig.6 shows the result that we got from world coordinate. Fig. 7 shows Position estimation by world co-ordinate system and orientation estimation.

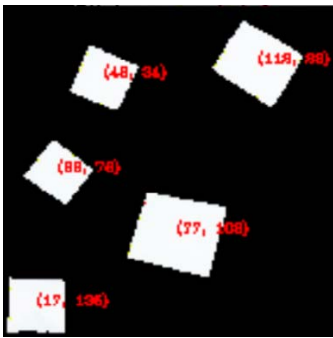


Fig. 6. World coordinate

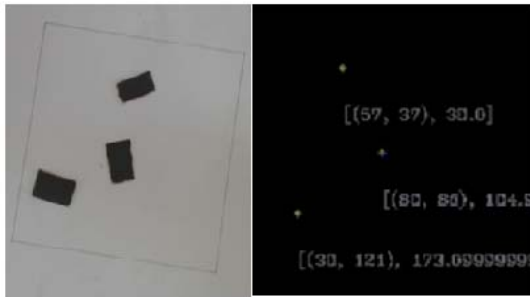


Fig. 7. Position estimation by world co-ordinate system and orientation estimation

VI. CONCLUSION

Most of the methods of visual grasping of objects are using technique of camera mounted on a robot. In this work, camera is mounted not on robot, but independently above the area of

interest. For that purpose there is need to convert camera co-ordinate into world co-ordinate. While performing this transformation, projective geometry and properties of 2D transformation such as scaling, rotation, translation plays an important role. The position of object is nothing but X,Y co-ordinate of centroid. Using socket communication we have send this coordinate to controller and it will take action so that respective servo motors for picking an object and placing it into defined place. The main objective is to apply it to industry problem.

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